*Course:* Functional methods in polymer statistical mechanics and stochastic processes
*Professor:* Prof. Enzo Orlandini

# Feynman-Kac formula to solve Schrödinger's equation and the derivation of the path integral

*Candidate:* Alice Magnanini (2091045)

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**Abstract**

In this paper we initially present the Feynman-Kac formula as a bridge between stochastic calculus and the Schrödinger equation. After recalling the construction of Brownian motion, the Wiener measure, and Itô formula, we show that a classical solution of a class of backward parabolic PDEs with potential can be represented as the expectation of a functional of an Itô process and the potential. Through the analytic continuation of the semigroup $e^{-sH}$ to the imaginary axis, $s = it/\hbar$, this yields a solution of the Schrödinger equation. This framework allows us to compute the propagator of a non-relativistic particle in a potential both as a Feynman-Kac solution of the Schrödinger equation and as a Feynman path integral, thereby clarifying the relation between Feynman's heuristic formulation of the path integral and its probabilistic counterpart. We conclude providing two examples: a worked computation of the Feynman-Kac solution to the Schrödinger equation of a free non-relativistic particle with an initial Gaussian wave packet; and the Black-Scholes formula of financial mathematics for the pricing of European options, computed with the Feynman-Kac formula and considering a geometric Brownian motion as driving process.

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# Contents

**Abstract** **1**

**References** **1**

**1 Introduction** **3**

**2 Probabilistic setting** **3**
2.1 Rigorous construction of Brownian motion . . . . . 3
2.2 Equivalent characterizations of Brownian motion . . . . . 5
2.3 Canonical representation of Brownian motion and the Wiener measure . . . . . 5

**3 Itô formula** **6**

**4 Feynman-Kac formula** **7**

**5 Solution to Schrödinger equation using Feynman-Kac formula** **8**

**6 Derivation of the path integral** **9**
6.1 Propagator as Feynman-Kac solution of Schrödinger equation . . . . . 9
6.2 The propagator of the free non-relativistic particle . . . . . 10
6.3 Propagator as Feynman's path integral . . . . . 10

**7 Examples** **11**
7.1 The free particle ($V = 0$) . . . . . 11
7.2 Black-Scholes PDE . . . . . 12

# 1 Introduction

The Feynman-Kac formula links two apparently distant worlds: the expectation of functionals of random paths and the solutions of linear parabolic PDEs. It originates in the work of Feynman and Kac in the late 1940s. Feynman proposed to write the solution of the Schrödinger equation as a "sum over all paths" (i.e., the *path integral*), whereas Kac observed that, after passing to imaginary time, the oscillating weights become killing factors and the formal path integral becomes a rigorous integral with respect to the Wiener measure.

The Schrödinger equation is not parabolic, so the Feynman–Kac formula does not apply to it directly. The strategy followed in this paper is the following: we first solve the real-valued equation $\partial_s u = -Hu$ by means of the Feynman-Kac formula, and we then recover the complex-valued solution by analytic continuation of the semigroup $e^{-sH}$ from the half-line $s > 0$ to the imaginary axis, $s = it/\hbar$. The spectral theorem for the self-adjoint operator $H = -\frac{1}{2}\frac{\hbar^2}{m}\partial_{xx}^2 + V$ makes this continuation precise: the map $\zeta \mapsto e^{-\zeta H}$ is holomorphic on $\{\mathrm{Re}\,\zeta > 0\}$, and on its boundary it reduces to the unitary group generated by the Hamiltonian.

The same construction allows us to revisit Feynman's original formulation of the path integral. Writing the solution of the Schrödinger equation in terms of its propagator, i.e., the function $K(x,t;x',t')$ that determines the probability for the system to go from the state $x'$ at time $t'$ to the state $x$ at time $t$, we compute it in two different ways. First, as a Feynman-Kac expectation, which we evaluate explicitly in the case of the free particle. Second, as Feynman's path integral, which we obtain by splitting the time interval into $n$ sub-intervals, applying the Trotter product formula, and inserting a completeness relation at each time step. In the limit $n \to \infty$, the resulting finite-dimensional integrals converge to a "sum over paths" weighted by $e^{iS[x]/\hbar}$, where $S[x]$ is the classical action on $\mathbf{x}$. Comparing the two characterizations shows that the Feynman-Kac representation and the time-sliced path integral define the same propagator. In this sense, the Feynman-Kac formula provides a probabilistic framework in which Feynman's heuristic construction can be understood: the finite-dimensional integrals are well defined and converge to the propagator, while the continuum notation $\int_{\mathbf{x}(t')=x'}^{\mathbf{x}(t)=x} \mathcal{D}[\mathbf{x}]\, e^{iS[\mathbf{x}]/\hbar}$ remains a formal shorthand for this limit.

We conclude showing that besides the Schrödinger equation, the same formula solves other PDEs, among which the Black-Scholes PDE of financial mathematics, where the driving process is not simply a rescaled Brownian motion, but instead a geometric Brownian motion. We use it as a last example, not tied to the quantum-mechanical setting.

**Organization of the paper.** Section 2 collects the probabilistic preliminaries: the construction of Brownian motion through Gaussian spaces, its equivalent characterizations and the Wiener measure; the material follows [Hof26]. Section 3 recalls the notions of Itô process and states Itô formula, following [Øks03]. Section 4 states and proves the Feynman-Kac formula for the backward PDE with potential; the statement and the existence part follow [Øks03]. Section 5 derives the solution of the Schrödinger equation from the Feynman-Kac formula through the time inversion and the analytic continuation of $e^{-sH}$. Section 6 introduces the propagator and derives it in two ways: as a Feynman-Kac solution of the Schrödinger equation, with the explicit computation for the free particle, and as Feynman's path integral, obtained through the Trotter product formula and time-slicing. Finally, Section 7 collects two worked examples: the free particle and the Black-Scholes PDE.

# 2 Probabilistic setting

## 2.1 Rigorous construction of Brownian motion

It is convenient to introduce the language of *Gaussian spaces* and *Gaussian processes* that will allow us to easily manipulate families of Gaussian random variables.

**Definition 1.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A *Gaussian space* is a closed vector subspace of $L^2(\Omega, \mathcal{F}, \mathbb{P})$ all of whose elements are centered Gaussian random variables.

**Definition 2.** Let $X = (X_t)_{t\in T}$ be a family of random variables. We say that $X$ is a *Gaussian process* if one of the following two conditions hold:

(i) $\mathrm{span}\{X_t,\ t \in T\}$ (i.e., the closure in $L^2$ of the vector space generated by the $X_t$'s) is a Gaussian space;
(ii) every finite combination of the $X_t$'s is a centered Gaussian random variable.

We now present a rigorous method to construct Gaussian processes indexed by an arbitrary real, separable Hilbert space $H$. The Brownian motion will emerge as canonical example associated with $H = L^2(\mathbb{R}_+, dt)$ (where $\mathbb{R}_+ := \mathbb{R}_{\geq 0}$ and $dt$ is the Lebesgue measure on $R_+$).

**Theorem 1.** *Given a real, separable Hilbert space $H$, there exists a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ carrying a family $(X(h),\ h \in H)$ of centered Gaussian random variables such that*

$$\forall h, k \in H, \qquad \mathbb{E}[X(h)X(k)] = \langle h, k\rangle_H, \tag{1}$$

*and the map $X : H \to L^2(\Omega, \mathcal{F}, \mathbb{P}),\ h \mapsto X(h)$ is linear.*

*Proof.* We treat separately the finite- and infinite-dimensional cases.

Case $\dim H < +\infty$. Let $(e_1, \dots, e_n)$ be an orthonormal basis of $H$ and let $(\xi_1, \dots, \xi_n)$ be i.i.d. RVs $\sim \mathcal{N}(0,1)$ on some $(\Omega, \mathcal{F}, \mathbb{P})$. Define, for $H \ni h = \sum_{i=1}^n \langle h, e_i\rangle_H\, e_i$,

$$X(h) := \sum_{i=1}^n \langle h, e_i\rangle_H\, \xi_i. \tag{2}$$

Then $\mathbb{E}[X(h)X(k)] = \sum_{i,j=1}^n \langle h, e_i\rangle_H \langle k, e_j\rangle_H \mathbb{E}[\xi_i\xi_j] = \sum_{i=1}^n \langle h, e_i\rangle_H \langle k, e_i\rangle_H = \langle h, k\rangle_H$, as required.

Case $\dim H = +\infty$. Take $(\Omega, \mathcal{F}, \mathbb{P})$ rich enough to support a sequence $(\xi_i)_{i\geq 1}$ of i.i.d. RVs $\sim \mathcal{N}(0,1)$, and let $(e_i)_{i\geq 1}$ be an Hilbert basis of $H$. For $H \ni h = \sum_{i=1}^{+\infty} \langle h, e_i\rangle_H\, e_i$, define:

$$X(h) := \sum_{i=1}^{+\infty} \langle h, e_i\rangle_H\, \xi_i; \tag{3}$$

note that this series is well defined as it converges in $L^2(\Omega, \mathcal{F}, \mathbb{P})$; indeed, its partial sums form a Cauchy sequence, since

$$\mathbb{E}\left[\left(\sum_{i=n}^m \langle h, e_i\rangle_H\, \xi_i\right)^2\right] = \sum_{i=n}^m \langle h, e_i\rangle_H^2 \xrightarrow{n,m\to+\infty} 0, \tag{4}$$

the tail of the convergent series $\sum_{i=1}^{+\infty} \langle h, e_i\rangle_H^2 = |h|_H^2$. It is then easy to check that $\mathbb{E}[(X(\alpha h + \beta k) - \alpha X(h) - \beta X(k))^2] = 0$, i.e., $h \mapsto X(h)$ is linear. □

**Remark 1.** The map $X : H \to L^2(\Omega, \mathcal{F}, \mathbb{P}),\ h \mapsto X(h)$ is an isometry: indeed,

$$\|X(h) - X(k)\|_2^2 = \mathbb{E}[|X(h) - X(k)|^2] = \mathbb{E}[X(h)^2 - 2X(h)X(k) + X(k)^2] = |h|_H^2 - 2\langle h, k\rangle_H + |k|_H^2 = |h-k|_H^2. \tag{5}$$

**Remark 2.** If $(h_n)_{n\geq 0} \in H$ and $h_n \xrightarrow{H} h$ then $X(h_n) \xrightarrow{L^2} X(h)$; in particular, $(X(h), h \in H)$ is a Gaussian space, and

$$X(h) = \lim_{\substack{n\to+\infty \\ L^2}} \sum_{i=0}^n \langle h, e_i\rangle_H\, \xi_i \sim \mathcal{N}\left(0, \lim_{n\to+\infty} \langle h, e_i\rangle_H^2\right) = \mathcal{N}\left(0, |h|_H^2\right). \tag{6}$$

**Remark 3.** If $\langle h, f\rangle_H = 0$ (i.e., $f, h$ are orthogonal in $H$) then $X(f) \perp\!\!\!\perp X(h)$ (i.e., $X(f), X(h)$ are independent in $L^2(\Omega, \mathcal{F}, \mathbb{P})$); indeed, using linearity of $X$ and the known formula for the characteristic function of normal distributions, we can write:

$$\varphi_{X(f)+X(h)}(\theta) = \varphi_{X(f+h)}(\theta) = \exp\left(-\frac{|f+h|_H^2}{2}\theta^2\right) \tag{7}$$

and

$$\varphi_{X(f)}(\theta)\varphi_{X(h)}(\theta) = \exp\left(-\frac{|f|_H^2}{2}\,\theta^2\right)\exp\left(-\frac{|h|_H^2}{2}\,\theta^2\right) = \exp\left(-\frac{|f|_H^2 + |h|_H^2}{2}\,\theta^2\right), \tag{8}$$

which coincide $\forall \theta \in \mathbb{R}$ since the orthogonality assumption implies $|f+h|_H^2 = |f|_H^2 + |h|_H^2$. Note that the same result holds for every finite collection of pairwise orthogonal functions $h_1, \dots, h_p \in H$.

**Definition 3.** When $H = L^2(T, \mathcal{G}, \mu)$ for some $\sigma$-finite measure $\mu$ on the measurable space $(T, \mathcal{G})$, then $X$ given by Theorem 1 is called *Gaussian measure of intensity $\mu$*.

**Remark 4.** In the particular case $f = \mathbb{1}_A$ where $A \in \mathcal{G}$, $\mu(A) < +\infty$, one writes $X(A) := X(\mathbb{1}_A)$, so that

$$X(A) = X(\mathbb{1}_A) \sim \mathcal{N}\left(0, |\mathbb{1}_A|_H^2\right) = \mathcal{N}\left(0, \mu(A)\right). \tag{9}$$

Moreover, if $A = \sum_h A_n$ is a countable disjoint union with $\mu(A_n) < +\infty\ \forall n$, then $\mu(A) = \sum_n \mu(A_n)$ and $X(A)$ can be written as a converging series in $L^2$: $X(A) = \sum_n X(A_n)$.

**Definition 4.** Take $T = \mathbb{R}_+$, $\mathcal{G} = \mathcal{B}(\mathbb{R}_+)$ and $\mu = dt$ the Lebesgue measure on $(\mathbb{R}_+, \mathcal{B}(\mathbb{R}_+))$. Then the process $B = (B_t)_{t\geq 0}$ defined as:

$$B_t := X([0,t]) = X(\mathbb{1}_{[0,t]}), \qquad t \in \mathbb{R}_+ \tag{10}$$

is a Gaussian process, and in particular we call it *Brownian motion* (BM).

**Remark 5.** Given a Brownian motion constructed as in Definition 4, it is not entirely straightforward that for each $\omega \in \Omega$, then $t \mapsto B_t(\omega)$ (a "path" or "trajectory") is a (at least) a measurable function. Indeed, depending on how $B$ is constructed, it may not be true for some $\omega \in \Omega$. However, thanks to Kolmogorov's continuity criterion (see [Hof26, Theorem 18]), up to a suitable modification one may always assume that the Brownian motion has continuous paths. From now on, whenever we say *Brownian motion* we will always refer, tacitly, to this continuous modification.

## 2.2 Equivalent characterizations of Brownian motion

The construction of Brownian motion given in Definition 4 is relatively simple, but rather abstract and not particularly convenient to work with. Fortunately, there are several equivalent characterizations of Brownian motion on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ that provide more convenient and intuitive ways of working with it. We will not prove the equivalence of the following definitions; one can find them in [Hof26].

**Karhunen-Loève characterization.** Let $\xi_0, \xi_1, \dots$ be independent and $\mathcal{N}(0,1)$-distributed random variables on $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$. Then,

$$B_t = \sum_{n=0}^{+\infty} \frac{\sqrt{2}\sin\left((n+\frac{1}{2})\pi t\right)}{\left(n+\frac{1}{2}\right)\pi}\,\xi_n, \qquad 0 \leq t \leq 1, \tag{11}$$

and $B = (B_t)_{t\geq 0}$ obtained by gluing i.i.d. copies of the random variable defined in (11) for the unitary interval, is a Brownian motion on $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$.

*Remark.* Note that the Karhunen-Loève characterization straightforwardly arises from Definition 4 if we take $(e_n)_{n\geq 0} := \left(s \mapsto \sqrt{2}\cos((n+\frac{1}{2})\pi s)\right)_{n\geq 0}$ as the orthonormal basis of $L^2([0,1])$.

**Characterization by independent and stationary increments.** A stochastic process $B = (B_t)_{t\geq 0}$ is a Brownian motion on $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ if:

(i) it is adapted to the filtration $(\mathcal{F}_t)_{t\geq 0}$ and $B_0 = 0$ a.s.;
(ii) $t \mapsto B_t$ is continuous a.s. (i.e., $\mathbb{P}(\{\omega \in \Omega : t \mapsto B_t(\omega) \text{ is continuous}\}) = 1$, that is, $B$ has continuous trajectories a.s.);
(iii) Stationary increments: $B_t - B_s \sim \mathcal{N}(0, t-s) \quad \forall\, 0 \leq s \leq t$;
(iv) Independence of increments: $B_{t_1}, B_{t_2} - B_{t_1}, \dots, B_{t_n} - B_{t_{n-1}}$ are independent $\forall\, 0 \leq t_1 < \cdots < t_n$.

*Remark.* (i)-(iv) can be trivially deduced from Definition 4 just using the properties of the linear isometry $X$.

**Characterization as Gaussian process.** A stochastic process $B = (B_t)_{t\geq 0}$ is a Brownian motion on $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ if:

(i) $B$ is a Gaussian process on $(\Omega, \mathcal{F}, \mathbb{P})$ adapted to $(\mathcal{F}_t)_{t\geq 0}$;
(ii) $t \mapsto B_t$ is continuous a.s. (i.e., $\mathbb{P}(\{\omega \in \Omega : t \mapsto B_t(\omega) \text{ is continuous}\}) = 1$, that is, $B$ has continuous trajectories a.s.);
(iii) Mean process: $\mathbb{E}[B_t] = 0\ \forall t \geq 0$;
(iv) Covariance structure: $\mathbb{E}[B_t B_s] = \min(s,t)\ \forall t, s \geq 0$.

## 2.3 Canonical representation of Brownian motion and the Wiener measure

Exploiting the existence of a continuous modification of Brownian motion, we can construct it on the space of continuous function; this will allow us to work always in the same general setting (the *Wiener space*), rather than assuming each time the existence of an arbitrary probability space $(\Omega, \mathcal{F}, \mathbb{P})$ carrying a Brownian motion.

Let $C(\mathbb{R}_+, \mathbb{R})$ denote the space of continuous $\mathbb{R}$-valued functions defined on $\mathbb{R}_+$. Given a continuous Brownian motion $B$ on an arbitrary probability space $(\Omega, \mathcal{F}, \mathbb{P})$, one may regard $B$ as a single random variable taking values in $C(\mathbb{R}_+, \mathbb{R})$, via the map

$$\psi : \Omega \longrightarrow C(\mathbb{R}_+, \mathbb{R}), \qquad \omega \mapsto \psi(\omega) := (t \mapsto B_t(\omega)). \tag{12}$$

To make sense of the measurability of this map, we equip $C(\mathbb{R}_+, \mathbb{R})$ with the smallest $\sigma$-field, denoted $\mathcal{C}$, making all the coordinate (or canonical) maps

$$X_t : C(\mathbb{R}_+, \mathbb{R}) \longrightarrow \mathbb{R}, \qquad w \mapsto X_t(w) := w(t) \tag{13}$$

$(\mathcal{C}, \mathcal{B}(\mathbb{R}))$-measurable, $\forall t \geq 0$; in other words, $\mathcal{C} := \sigma(X_t,\ t \geq 0)$. This makes $\psi$ $(\mathcal{F}, \mathcal{C})$-measurable: indeed, $(B_s(\omega))_{s\geq 0} = (X_s(t \mapsto B_t(\omega)))_{s\geq 0} = (X_s(\psi(\omega)))_{s\geq 0}$, so by continuity of $\psi(\omega)$ for *all* $\omega$ we can say

$$\psi(\omega) = \bigcap_{s\geq 0} X_s^{-1}(B_s(\omega)) = \bigcap_{s\in\mathbb{Q}\cap[0,+\infty)} X_s^{-1}(B_s(\omega)), \tag{14}$$

which is countable intersection of $\mathcal{C}$-measurable sets (since coordinate maps are measurable), hence it is $\mathcal{C}$-measurable.

*Remark.* Note that it is possible to show that $\mathcal{C}$ coincides with the Borel $\sigma$-field on $C(\mathbb{R}_+, \mathbb{R})$ associated with the topology of uniform convergence on compact sets.

**Definition 5.** Let $B$ be a Brownian motion on $(\Omega, \mathcal{F}, \mathbb{P})$ and let $\psi$ be the associated $\mathcal{C}$-measurable path map defined in (12). The *Wiener measure* is the push-forward of $\mathbb{P}$ under $\psi$:

$$\mathbb{W} := \psi \# \mathbb{P}, \tag{15}$$

which is a probability measure on $(C(\mathbb{R}_+, \mathbb{R}), \mathcal{C})$.

**Definition 6.** The probability space $(C(\mathbb{R}_+, \mathbb{R}), \mathcal{C}, \mathbb{W})$ is called *Wiener (probability) space*, or *canonical probability space*.

**Theorem 2.** *Under $\mathbb{W}$, the canonical process of the coordinate (canonical) maps $(X_t)_{t\geq 0}$, living directly on $(C(\mathbb{R}_+, \mathbb{R}), \mathcal{C})$, is itself a Brownian motion.*

*Proof.* For any $0 = t_0 < t_1 < \cdots < t_n$ and Borel sets $A_0, \ldots, A_n \in \mathcal{B}(\mathbb{R})$,

$$\begin{aligned}
&\mathbb{W}\left(X_{t_0} \in A_0, \ldots, X_{t_n} \in A_n\right) = \\
&\quad = \mathbb{W}(\{w \in C(\mathbb{R}_+, \mathbb{R}) \ : \ X_{t_0}(w) = w(t_0) \in A_0\}, \ldots, \{w \in C(\mathbb{R}_+, \mathbb{R}) \ : \ X_{t_n}(w) = w(t_n) \in A_n\}) \\
&\quad = \mathbb{P}(\{\omega \in \Omega \ : \ \psi(\omega)(t_0) \in A_0\}, \ldots, \{\omega \in \Omega \ : \ \psi(\omega)(t_n) \in A_n\}) \\
&\quad = \mathbb{P}(\{\omega \in \Omega \ : \ B_{t_0}(\omega) \in A_0\}, \ldots, \{\omega \in \Omega \ : \ B_{t_n}(\omega) \in A_n\}) \\
&\quad = \mathbb{1}_{A_0}(0) \int_{A_1 \times \cdots \times A_n} \frac{dx_1 \ldots dx_n}{(2\pi)^{n/2}\sqrt{t_1(t_2 - t_1)\ldots(t_n - t_{n-1})}} \exp\left(-\frac{1}{2}\sum_{i=1}^{n} \frac{(x_i - x_{i-1})^2}{t_i - t_{i-1}}\right),
\end{aligned} \tag{16}$$

where the last equality is true simply by the joint Gaussian density of the increments of any Brownian motion $B$. As a consequence, $\mathbb{W}$ and $(X_t)_{t\geq 0}$ do not depend on the particular Brownian motion $B$ used in the construction, as they are entirely characterized by the finite-dimensional distributions in (16). □

**Remark 6.** The last assertion relies on the fact that if $\mathbb{P}, \mathbb{Q}$ are two probability measures on $(C(\mathbb{R}_+, \mathbb{R}), \mathcal{C})$ associated to two continuous processes $Z \sim \mathbb{P}, \tilde{Z} \sim \mathbb{Q}$ and $Z, \tilde{Z}$ have same finite-dimensional distributions, then $\mathbb{P} = \mathbb{Q}$. This is not completely trivial: one can show it using that two such measures coincide if and only if they coincide on a $\pi$-system, and that cylinder sets form, indeed, a $\pi$-system for $(C(\mathbb{R}_+, \mathbb{R}), \mathcal{C})$.

# 3 Itô formula

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ be a filtered probability space carrying a standard Brownian Motion $B$. Consider the one-dimensional stochastic differential equation (SDE):

$$dX_s = b(s, X_s)ds + \sigma(s, X_s)dB_s, \quad X_0 = x, \tag{17}$$

where $b, \sigma : [0, T] \times \mathbb{R} \longrightarrow \mathbb{R}$ are, respectively, *drift* and *volatility* coefficients, and $T \in [0, +\infty]$ is the horizon (if we have, $T = +\infty$, i.e., infinite horizon, then we consider the domain $[0, +\infty) \times \mathbb{R}$).

**Definition 7.** An *Itô process* or *stochastic integral* is any stochastic process on $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ adapted to $(\mathcal{F}_t)_{t\geq 0}$ which can we written in the form $X = (X_t)_{t\geq 0}$, with

$$X_t = x + \int_0^t b(s, X_s)ds + \int_0^t \sigma(s, X_s)dB_s, \tag{18}$$

(or Eq. (17) as shorthand notation), where $b, \sigma$ are progressively measurable and

$$\mathbb{P}\left(\int_0^t |b(s, \omega)|ds < +\infty \ \forall t \geq 0\right) = 1, \qquad \mathbb{P}\left(\int_0^t \sigma(s, \omega)^2 ds < +\infty \ \forall t \geq 0\right) = 1. \tag{19}$$

If moreover $b, \sigma$ do not depend on time, then the unique strong solution to Eq. (17) is called a (time-homogeneous) *Itô diffusion*.

**Theorem 3** (Existence and uniqueness of strong solution)**.** *If $b, \sigma : [0, T] \times \mathbb{R} \longrightarrow \mathbb{R}$ are Lipschitz in the space variable and with linear growth, then Eq.* (17) *has a unique strong solution for any initial condition $X_t = x$ with $(t, x) \in [0, T] \times \mathbb{R}$.*

*Proof.* See [Øks03, Theorem 5.2.1]. □

**Theorem 4** (Itô formula)**.** *For any $h \in C^{1,2}([0,T]\times\mathbb{R})$ and Itô process $X$, the stochastic process $Y$ defined as $Y_t := h(t, X_t)\ \forall t \geq 0$ is again an Itô process and*

$$dY_t = dh(t,X_t) = \partial_t h(t,X_t)dt + \partial_x h(t,X_t)dX_t + \frac{1}{2}\partial^2_{xx}h(t,X_t)d\langle X\rangle_t, \qquad Y_0 = h(0,X_0). \tag{20}$$

A proof of the Itô formula can be found in [Øks03, Theorem 4.1.2]. Note that, for a Itô process $X$ satisfying Eq. (17), then the quadratic variation is $d\langle X\rangle_t = \sigma^2(t,X_t)dt$ and consequently Itô formula yields:

$$dh(t,X_t) = \left(\partial_t h(t,X_t) + b(t,X_t)\partial_x h(t,X_t) + \frac{1}{2}\sigma^2(t,X_t)\partial^2_{xx}h(t,X_t)\right)dt + \sigma(t,X_t)\partial_x h(t,X_t)dB_t. \tag{21}$$

**Remark 7.** It is possible to generalize the Itô formula to $\mathbb{R}^d$-valued Itô diffusions: in this case, $b, \sigma$ are $\mathbb{R}^d$-valued and Eq. (20),(21) become, respectively, $\forall h \in C^2(\mathbb{R}^d, \mathbb{R})$:

$$dh(X_t) = \sum_{i=1}^{d}\partial_{x_i}h(X_t)dX_t^{(i)} + \frac{1}{2}\sum_{i,j=1}^{d}\partial^2_{x_ix_j}h(X_t)d\langle X^{(i)}, X^{(j)}\rangle_t = \tag{22}$$

$$= \left(\nabla h(X_t)\cdot b(X_t) + \frac{1}{2}\mathrm{Tr}\left(\sigma(X_t)^\top D^2 h(X_t)\sigma(X_t)\right)\right)dt + \nabla h(X_t)\cdot\sigma(X_t)dB_t. \tag{23}$$

# 4 Feynman-Kac formula

> *It is probably safe to say that I am better and more widely known for being the K in the F-K formula than for anything else I have done during my scientific career...*
>
> *Mark Kac, 1984 [Kac87, p. 115]*

Consider the *backward Feynman-Kac PDE*:

$$\begin{cases} \partial_t u(t,x) + b(t,x)\partial_x u(t,x) + \frac{1}{2}\sigma^2(t,x)\partial^2_{xx}u(t,x) - V(t,x)u(t,x) + f(t,x) = 0, & \forall (t,x)\in[0,T)\times\mathbb{R}, \\ u(T,x) = \psi(x), & \forall x\in\mathbb{R}, \end{cases} \tag{24}$$

where $b, \sigma, f : [0,T]\times\mathbb{R} \longrightarrow \mathbb{R}$ are known coefficients (respectively, drift, volatility and source), $\psi : \mathbb{R} \longrightarrow \mathbb{R}$ is a known terminal condition, $V : [0,T]\times\mathbb{R} \longrightarrow \mathbb{R}$ is a known potential bounded from below, and $u : [0,T]\times\mathbb{R} \longrightarrow \mathbb{R}$ is the unknown function. The goal is to provide a characterization of the (unique) classical solution to Eq. (24) under certain regularity and integrability assumptions.

**Theorem 5** (Feynman-Kac theorem)**.** *Assume that a solution $u \in C^{1,2}([0,T)\times\mathbb{R}) \cap C([0,T]\times\mathbb{R})$ to Eq. (24) exists, and that $u, f$ and $\sigma\partial_x u$ satisfy a polynomial growth condition in $x$. Then $\forall (t,x)\in[0,T]\times\mathbb{R}$,*

$$u(t,x) = \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_t^T V(r,X_r)dr}\psi(X_T) + \int_t^T e^{-\int_t^s V(r,X_r)dr}f(s,X_s)ds \Big| X_t = x\right], \tag{25}$$

*where $X = (X_t)_{t\geq 0}$ is an Itô process on some filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{Q})$ satisfying the SDE:*

$$dX_t = b(t,X_t)dt + \sigma(t,X_t)dB_t^{\mathbb{Q}}, \qquad t\in[0,T], \tag{26}$$

*with $b, \sigma$ Lipschitz in $x$ and with linear growth, and $B^{\mathbb{Q}} = (B_t^{\mathbb{Q}})_{t\geq 0}$ a standard Brownian motion under $\mathbb{Q}$. In particular, under these assumptions the solution to Eq. (24) is unique.*

**Remark 8.** The original statement of Feynman-Kac theorem establishes both existence and uniqueness, and provides a probabilistic characterization of the unique solution (Eq. (25)), which is the famous so-called *Feynman-Kac formula*. We will show that whenever the solution exists, then it must satisfy the Feynman-Kac formula; proof of existence and uniqueness is not presented here (for reference, see [Øks03, Theorem 8.2.1]).

*Proof (partial).* Let $g(t,s) := e^{-\int_t^s V(r,X_r)dr}$, where $X$ is the stochastic process solving the SDE (26); then its differential satisfies:

$$dg(t,s) = -V(s,X_s)g(t,s)ds. \tag{27}$$

Define now the stochastic process $Y = (Y_\tau)_{\tau\in[t,T]}$ as:

$$Y_\tau := g(t,\tau)u(\tau,X_\tau) + \int_t^\tau g(t,s)f(s,X_s)ds. \tag{28}$$

We can compute $Y$ explicitly at boundary times $\tau \in \{t, T\}$:

$$Y_t = u(t, X_t), \qquad Y_T = g(t,T)\psi(X_T) + \int_t^T g(t,s) f(s, X_s) ds; \tag{29}$$

in particular, note that $Y_T$ is exactly the quantity inside the conditional expectation in Eq. (25). If we show that $Y$ is a martingale under $\mathbb{Q}$, then we would have $\forall (t,x) \in [0,T] \times \mathbb{R}$:

$$\mathbb{E}^{\mathbb{Q}}[Y_T | X_t = x] = \mathbb{E}^{\mathbb{Q}}[Y_T | Y_t = u(t,x)] = \mathbb{E}^{\mathbb{Q}}[Y_T | \mathcal{F}_t]\Big|_{Y_t = u(x,t)} = Y_t\Big|_{Y_t = u(t,x)} = u(t,x). \tag{30}$$

We now prove that $Y$ is a $\mathbb{Q}$-martingale. Apply Itô formula to $u$, which was assumed to be in $C^{1,2}$:

$$du(\tau, X_\tau) = \left(\partial_\tau u + b\, \partial_x u + \frac{1}{2}\sigma^2\, \partial^2_{xx} u\right)(\tau, X_\tau)\, d\tau + (\sigma\, \partial_x u)(\tau, X_\tau)\, dB^{\mathbb{Q}}_\tau. \tag{31}$$

Using this result, compute the differential of $Y_\tau$ with the stochastic chain rule:

$$dY_\tau = g(t,\tau)\, du(\tau, X_\tau) + u(\tau, X_\tau)\, dg(t,\tau) + d\langle g, u\rangle_\tau + g(t,\tau) f(\tau, X_\tau)\, d\tau, \tag{32}$$

where $d\langle g, u\rangle_\tau = 0$ since $\langle d\tau, d\tau\rangle = 0$ and $\langle d\tau, dB^{\mathbb{Q}}_\tau\rangle = 0$ . Substitute the differentials with their explicit formulas (27), (31), so that we get:

$$dY_\tau = g(t,\tau)\left(\partial_\tau u + b\, \partial_x u + \frac{1}{2}\sigma^2\, \partial^2_{xx} u - Vu + f\right)(\tau, X_\tau)\; d\tau + g(t,\tau)\,(\sigma\, \partial_x u)(\tau, X_\tau)\, dB^{\mathbb{Q}}_\tau. \tag{33}$$

Recall that $u$ solves the Feynman-Kac PDE (24) for every $(t,x) \in [0,T) \times \mathbb{R}$, so in particular for $(\tau, X_\tau)$; therefore, the drift term of $Y$ is null, making it a local martingale under $\mathbb{Q}$. But $V$ is bounded from below, which implies $g$ bounded, and together with polynomial growth assumptions, we conclude that the integrand is in $L^2$, so $Y$ becomes a true martingale. □

**Remark 9.** Feynman-Kac formula naturally admits a $d$-dimensional generalization, which yields a function $u : [0,T] \times \mathbb{R}^d \to \mathbb{R}$ solving the following PDE:

$$\begin{cases} \partial_t u + \sum_{i=1}^d b_i(t,x)\, \partial_{x_i} u + \frac{1}{2}\sum_{i,j=1}^d \left(\sigma\sigma^\top\right)_{ij}(t,x)\, \partial^2_{x_i x_j} u - V(t,x)\, u + f(t,x) = 0, & \forall (t,x) \in [0,T) \times \mathbb{R}^d, \\ u(T,x) = \psi(x) \in \mathbb{R}, & \forall x \in \mathbb{R}^d. \end{cases} \tag{34}$$

# 5 Solution to Schrödinger equation using Feynman-Kac formula

For a single non-relativistic particle of mass $m$ in a potential $V$, Schrödinger equation reads:

$$i\hbar \partial_t \psi = -\frac{\hbar^2}{2m}\partial^2_{xx}\psi + V\psi, \qquad (t,x) \in [0, +\infty) \times \mathbb{R}. \tag{35}$$

Here, $\psi : [0, +\infty) \times \mathbb{R} \longrightarrow \mathbb{C}$ is a wave function, whose squared absolute value gives rise to a probability density function; indeed, $|\psi(t,x)|^2$ can be interpreted as the probability density of the system (in this case, the single non-relativistic particle) being in position $x$ at time $t$.

Our goal is to compute the solution $\psi$ to the Schrödinger equation (35) with initial condition $\psi(0, \cdot) = \psi_0$ by exploiting the Feynman–Kac characterization of solutions to backward PDEs of the form given in Eq. (24). For the sake of simplicity, one could choose to assume $m = \hbar = 1$; here, we decided to maintain the dependence on an arbitrary positive mass $m > 0$ and the reduced Planck constant $\hbar > 0$.

For an arbitrary horizon $T > 0$, consider the following function:

$$w : [0,T] \times \mathbb{R} \longrightarrow \mathbb{R}, \qquad w(\tau, x) := \mathbb{E}^{\mathbb{Q}}\left[\psi_0(\hbar m^{-1/2} B^{\mathbb{Q}}_T) e^{-\int_\tau^T V(\hbar m^{-1/2} B^{\mathbb{Q}}_r)\, dr} \;\middle|\; \hbar m^{-1/2} B^{\mathbb{Q}}_\tau = x\right]. \tag{36}$$

By Feynman-Kac theorem 5, $w$ is the unique solution to the Feynman-Kac PDE with potential $V$, terminal condition $\psi_0$, and constant coefficients $b = 0$, $\sigma = \hbar m^{-1/2}$, $f = 0$:

$$\begin{cases} \partial_\tau w + \frac{\hbar^2}{2m}\partial^2_{xx} w - Vw = 0, & \forall (\tau, x) \in [0,T) \times \mathbb{R}, \\ w(T,x) = \psi_0(x), & \forall x \in \mathbb{R}, \end{cases} \tag{37}$$

with driving process $dX_\tau = \hbar m^{-1/2} dB^{\mathbb{Q}}_\tau$, $\tau \in [0,T]$.

To convert the backward PDE into a forward PDE, we perform a time-inversion: let $s := T - \tau$ and consider $u : [0,T] \times \mathbb{R} \longrightarrow \mathbb{R}$ defined by $u(s,x) := w(T-s,x) = w(\tau,x)$, so that $\partial_s u(s,x) = \partial_s w(T-s,x) = \partial_\tau w(\tau,x)\frac{d\tau}{ds} = -\partial_\tau w(\tau,x)$ and $\partial^2_{xx} u(s,x) = \partial^2_{xx} w(\tau,x)$. We deduce that $u$ is solution to the following forward PDE:

$$\begin{cases} -\partial_s u + \frac{\hbar^2}{2m}\partial^2_{xx} u - Vu = 0, & \forall (s,x) \in (0,T] \times \mathbb{R}, \\ u(0,x) = w(T,x) = \psi_0(x), & \forall x \in \mathbb{R}. \end{cases} \tag{38}$$

It is convenient to rewrite (38) in operator form, i.e.,

$$\partial_s u = \frac{\hbar^2}{2m}\partial^2_{xx} u - Vu = -Hu, \qquad H := -\frac{\hbar^2}{2m}\partial^2_{xx} + V, \tag{39}$$

since this notation allows to write the solution $u$ as action of an exponential operator on $\psi_0$, naming:

$$u(s,x) = e^{-sH}\psi_0(x), \qquad \forall (s,x) \in [0,T] \times \mathbb{R}. \tag{40}$$

Note that $H$ is a self-adjoint operator, since it is sum of a scaled Laplacian (which is a self-adjoint operator) and a real, bounded from below multiplication operator (which does not influence self-adjointedness). Therefore, spectral theorem gives an integral representation of $H$, namely $H = \int_{\sigma(H)} \lambda \, d\mu(\lambda)$, where $\sigma(H)$ is the spectrum of $H$ and $\mu$ a projection-valued measure on $\sigma(H)$. This representation allows to write

$$e^{-sH} = \int_{\sigma(H)} e^{-s\lambda} \, d\mu(\lambda), \qquad s \geq 0 \tag{41}$$

so that now it is natural to define an extension to complex times:

$$\forall \zeta \in \mathbb{C} : \Re(\zeta) > 0, \qquad e^{-\zeta H} := \int_{\sigma(H)} e^{-\zeta\lambda} \, d\mu(\lambda). \tag{42}$$

It can be proved that $u(\zeta, x) := e^{-\zeta H}\psi_0(x)$ is the unique analytic extension in $\{\zeta \in \mathbb{C} : \Re(\zeta) > 0\}$ of $u(s,x) = e^{-sH}\psi_0(x)$, which was defined for real times $s \in [0,T]$ with $T$ arbitrary horizon, hence $\forall s \geq 0$. This result is known as *Wick rotation*, and it requires additional theoretical machinery, which lies beyond the scope of this presentation. For a general reference, see [Paz83, Chapter 2] and [Zee10, Section V.2].

Now, consider the wave function $\psi : [0,T] \times \mathbb{R} \longrightarrow \mathbb{R}$ defined by $\psi(t,x) := u(it\hbar^{-1}, x) = e^{-it\hbar^{-1}H}\psi_0(x)$ (note that here $it\hbar$ lies on $\partial\{\zeta \in \mathbb{C} : \Re(\zeta) > 0\}$); by spectral calculus, $\psi$ satisfies:

$$i\hbar\,\partial_t\psi(t,x) = i\hbar\,\partial_t u\left(it\hbar^{-1}, x\right) = i\hbar\,\partial_t\left(e^{-it\hbar^{-1}H}\psi_0(x)\right) = i\hbar\left(-\frac{iH}{\hbar}\right)e^{-it\hbar^{-1}H}\psi_0(x) = H\psi(t,x). \tag{43}$$

Hence, $\psi$ satisfies the Schrödinger equation (35) with initial condition $\psi(0,\cdot) = u(0,\cdot) = \psi_0$.

# 6 Derivation of the path integral

## 6.1 Propagator as Feynman-Kac solution of Schrödinger equation

Under Schrödinger notation, Eq. (35) and its solution can be rewritten as:

$$i\hbar\frac{d}{dt}|\psi\rangle = H|\psi\rangle, \qquad |\psi(t)\rangle = e^{-itH/\hbar}|\psi_0\rangle, \qquad \forall t \geq 0. \tag{44}$$

Note that $\forall\, 0 \leq t' \leq t$, it holds also $|\psi(t)\rangle = e^{-i(t-t')H/\hbar}\,|\psi(t')\rangle$. Introducing a completeness relation for the position eigenstates $\int dx'|x'\rangle\langle x'| = \mathbb{1}$, in the coordinate representation the solution takes the form:

$$\psi(t,x) = \langle x|\psi(t)\rangle = \int dx'\langle x|e^{-i(t-t')H/\hbar}|x'\rangle\langle x'|\psi(t')\rangle = \int dx' K(x,t;x',t')\psi(t',x'), \tag{45}$$

where we have introduced the *transition kernel* or *propagator*:

$$K(x,t;x',t') := \langle x|e^{-i(t-t')H/\hbar}|x'\rangle. \tag{46}$$

Hence $|\psi\rangle$ can be rewritten as function of $K$, and so derivatives of $\psi$ in Eq. (44) can be naturally switched into derivatives of $K$; so, $K$ satisfies a Schrödinger equation, namely:

$$i\hbar\frac{d}{dt}K(x,t;x',t') = HK(x,t;x',t') \tag{47}$$

where $x', t'$ are fixed and derivatives act on just $x, t$ coordinates. In addition, $K$ obeys to the initial condition:

$$\lim_{t\to t'} K(x,t;x',t') = \delta(x-x'). \tag{48}$$

By Feynman-Kac theorem, it is evident that we already have a strategy to compute $K$ explicitly:

$$K(x,t;x',t') = \mathbb{E}^{\mathbb{Q}}\left[\delta(X_T - x')\, e^{-\int_{T-s}^{T} V(r,X_r)dr}\Big| X_{T-s} = x\right]_{s=i(t-t')/\hbar}, \qquad dX_\cdot = \hbar m^{-1/2} dB^{\mathbb{Q}}_\cdot. \tag{49}$$

## 6.2 The propagator of the free non-relativistic particle

The dynamics of the propagator $K$ of the free non-relativistic particle dynamics is determined by:

$$i\hbar\partial_t K = -\frac{\hbar^2}{2m}\partial^2_{xx} K, \qquad \lim_{t\to t'} K(x,t;x',t') = \delta(x-x'), \tag{50}$$

which is a Feynman-Kac PDE with coefficients $b = 0$, $\sigma = \hbar m^{-1/2}$, $V = 0$, $f = 0$. Applying Feynman-Kac formula the same as we did for $\psi$ with, now, intial condition $\delta(\cdot - x')$, we get:

$$w(\tau, x) = \mathbb{E}^{\mathbb{Q}}\left[\delta\left(\hbar m^{-1/2} B^{\mathbb{Q}}_T - x'\right)\Big| \hbar m^{-1/2} B^{\mathbb{Q}}_\tau = x\right], \tag{51}$$

and by applying time-inversion, stationarity of increments of the Brownian motion, and traslation of the initial condition,

$$u(s,x) = \mathbb{E}^{\mathbb{Q}}\left[\delta\left(\hbar m^{-1/2} B^{\mathbb{Q}}_T - x'\right)\Big| \hbar m^{-1/2} B^{\mathbb{Q}}_s = x\right] = \mathbb{E}^{\mathbb{Q}}\left[\delta\left(x + \hbar m^{-1/2} B^{\mathbb{Q}}_s - x'\right)\right]. \tag{52}$$

Using that $B^{\mathbb{Q}}_s \sim \mathcal{N}(0,s)$ under $\mathbb{Q}$, we just integrate against the Gaussian distribution and obtain:

$$u(s,x) = \int \delta\left(x + \hbar m^{-1/2} y - x'\right) \frac{1}{\sqrt{2\pi s}} e^{-\frac{y^2}{2s}} dy = \sqrt{\frac{m}{2\pi s\hbar^2}}\, e^{-\frac{m(x'-x)^2}{2s\hbar^2}} \tag{53}$$

Substituting $s = i(t-t')/\hbar$ (here, we subtract $t'$ as our initial condition is now at time $t'$ and not at 0), we get:

$$K(x,t;x',t') = \sqrt{\frac{m}{2\pi i(t-t')\hbar^{-1}\hbar^2}}\, e^{-\frac{m(x'-x)^2}{2i(t-t')\hbar^{-1}\hbar^2}} = \sqrt{\frac{m}{2\pi i(t-t')\hbar}}\, e^{\frac{im(x-x')^2}{2(t-t')\hbar}}. \tag{54}$$

## 6.3 Propagator as Feynman's path integral

We now show how $K$ can be written in terms of Feynman's path integral, and compare this with the Feynman-Kac representation (49). Following the same route of Feynman's to construct the path integral, divide $[t', t]$ into $n$ sub-intervals of length $\Delta t/n = (t-t')/n$, and take conventionally $t_0 := t'$, $t_n := t$. Let $H = H_0 + V$, $H_0 = -\hbar^2\partial^2_{xx}/(2m)$. By Trotter product (see [JL00, Theorem 7.5.2]), we can write:

$$e^{-i\Delta t H/\hbar} = e^{-i\Delta t(H_0+V)/\hbar} = \lim_{n\to+\infty}\left(e^{-i\Delta t H_0/\hbar n} e^{-i\Delta t V/\hbar n}\right)^n. \tag{55}$$

Applying this to Eq. (46) and inserting $n-1$ times the completeness relation $\int dx_j |x_j\rangle\langle x_j| = \mathbb{1}$, we obtain:

$$K(x,t;x',t') := \lim_{n\to+\infty}\langle x|(e^{-i\Delta t H_0/\hbar n} e^{-i\Delta t V/\hbar n})^n|x'\rangle = \lim_{n\to\infty}\int dx_1\cdots dx_{n-1}\prod_{j=0}^{n-1}\langle x_{j+1}|e^{-i\Delta t H_0/\hbar n} e^{-i\Delta t V/\hbar n}|x_j\rangle, \tag{56}$$

where we take conventionally $x_0 := x'$, $x_n := x$. Note that $V$ is diagonal in the eigenstates base, so $\langle x_{j+1}|e^{-i\Delta t H_0/\hbar n} e^{-i\Delta t V/\hbar n}|x_j\rangle = \langle x_{j+1}|e^{-i\Delta t H_0/\hbar n}|x_j\rangle e^{-i\Delta t V(x_j)/\hbar n}$. We already computed the remaining $\langle\cdot|\cdot|\cdot\rangle$: it is the propagator of the free non relativistic particle, Eq. (54); hence, taking $A_n := \sqrt{mn/2\pi i\hbar\Delta t}$, we can write:

$$K(x,t;x',t') = \lim_{n\to+\infty}(A_n)^n \int dx_1\cdots dx_{n-1}\, e^{iS_n(\mathbf{x})/\hbar}, \tag{57}$$

where

$$S_n(\mathbf{x}) := \frac{m}{2}\sum_{j=0}^{n-1}\frac{\Delta t}{n}\left(\frac{x_{j+1}-x_j}{\Delta t/n}\right)^2 - \sum_{j=0}^{n-1}\frac{\Delta t}{n}V(x_j), \qquad \mathbf{x} \;:\; \mathbf{x}(t_j) = x_j \;\forall j = 0,\dots,n-1. \tag{58}$$

Note that $S_n(\mathbf{x})$ is just a Riemann sum approximation of the classical action of a particle moving along the broken line path $(x_j \rightsquigarrow x_{j+1})_{j=0}^{j=n-1} \xrightarrow[n\to+\infty]{} \mathbf{x}$, so

$$\lim_{n\to+\infty} S_n(\mathbf{x}) = \int_0^t ds \left(\frac{m}{2}\dot{\mathbf{x}}^2 - V(\mathbf{x})\right) =: S[\mathbf{x}]. \tag{59}$$

We are now ready to write $K$ as a path integral: at the limit $n \to +\infty$, we are just interested in the paths $\mathbf{x}$ such that $\mathbf{x}(t') = x'$ and $\mathbf{x}(t) = x$, since we are integrate all over the infinite intermediate positions $x_1, \dots, x_{n-1}$ and intervals of length $\Delta t/n \to 0$. Hence, we can effectively substitute $S_n(\mathbf{x})$ with $S[\mathbf{x}]$ at the limit, and *integrate over all such possible paths*:

$$K(x,t;x',t') = \lim_{n\to+\infty} (A_n)^n \int dx_1 \cdots dx_{n-1}\, e^{iS_n(\mathbf{x})/\hbar} =: \int_{\mathbf{x}(t')=x'}^{\mathbf{x}(t)=x} \mathcal{D}[\mathbf{x}] e^{iS[\mathbf{x}]/\hbar}, \tag{60}$$

We conclude that the propagator $K$ admits two complementary representations: the Feynman-Kac representation of Eq. (49), obtained from the Wiener measure through analytic continuation, and the time-sliced limit of Eq. (57), whose formal continuum limit is Feynman's path integral (60). Since both describe the same kernel, the Feynman-Kac approach gives a mathematically controlled interpretation of the path integral. We stress, however, that the symbol $\int_{\mathbf{x}(t')=x'}^{\mathbf{x}(t)=x} \mathcal{D}[\mathbf{x}] e^{iS[\mathbf{x}]/\hbar}$ is not an integral against a genuine measure (the oscillatory weight $e^{iS/\hbar}$ has no Wiener-type counterpart), and should be read as shorthand for the limit in Eq. (57).

# 7 Examples

## 7.1 The free particle ($V = 0$)

Consider a single non-relativistic free particle of mass $m$ in a potential-free space ($V = 0$). We would like to find its probability density on $[0, +\infty) \times \mathbb{R}$, assuming that at initial time $t = 0$ it is a Gaussian wave packet $\psi_0(z) := e^{-az^2}$, $a > 0$. A way to do so can be solving the associated Schrödinger equation through Feynman-Kac formula and find the amplitude $\psi$, which generates the probability density.

The Schrödinger equation associated to the system is:

$$\begin{cases} i\hbar\partial_t\psi = -\frac{\hbar^2}{2m}\partial_{xx}^2\psi, & \forall (t,x) \in [0,+\infty) \times \mathbb{R}, \\ \psi(0,x) = \psi_0(x) = e^{-ax^2} & a > 0,\ \forall x \in \mathbb{R}. \end{cases} \tag{61}$$

Fix an arbitrary horizon $T > 0$; then $\forall (\tau, x) \in [0,T] \times \mathbb{R}$,

$$w(\tau,x) = \mathbb{E}^{\mathbb{Q}}\left[e^{-a(\hbar m^{-1/2} B_T^{\mathbb{Q}})^2} \middle| \hbar m^{-1/2} B_\tau^{\mathbb{Q}} = x\right]. \tag{62}$$

Let $a' := a(\hbar m^{-1/2})^2 > 0$, $x' := x\hbar^{-1}m^{1/2} \in \mathbb{R}$; after performing time-inversion and exploiting stationarity of Brownian increments, we get, $\forall (s,x) \in [0,T] \times \mathbb{R}$:

$$u(s,x) = w(T-s,x) = \mathbb{E}^{\mathbb{Q}}\left[e^{-a'(B_T^{\mathbb{Q}})^2} \middle| B_{T-s}^{\mathbb{Q}} = x'\right] = \mathbb{E}^{\mathbb{Q}}\left[e^{-a'(x'+B_s^{\mathbb{Q}})^2}\right] \tag{63}$$

Now, $B_s^{\mathbb{Q}} \sim \mathcal{N}(0,s)$ under $\mathbb{Q}$, so we can expand the last formula by completing the square:

$$u(s,x) = \int_{-\infty}^{+\infty} e^{-a'(x'+y)^2} \cdot \frac{1}{\sqrt{2\pi s}} e^{-\frac{y^2}{2s}} dy = \frac{\exp\left(-a'x'^2 + \frac{2a'^2x'^2 s}{1+2a's}\right)}{\sqrt{2\pi s}} \int_{-\infty}^{+\infty} \exp\left(-\frac{1}{2}\frac{\left(y + \frac{2a'x's}{1+2a's}\right)^2}{\frac{s}{1+2a's}}\right) dy, \tag{64}$$

finally, we recognize the integral over $\mathbb{R}$ of a normal distribution of volatility $\frac{s}{1+2a's}$; so we get:

$$u(s,x) = \frac{\exp\left(-\frac{a'x'^2}{1+2a's}\right)}{\sqrt{2\pi s}} \cdot \sqrt{\frac{2\pi s}{1+2a's}} = (1+2a's)^{-1/2} \exp\left(-\frac{a'x'^2}{1+2a's}\right). \tag{65}$$

Now, we extend into $\mathbb{C}$: we get, $\forall (t,x) \in [0,T] \times \mathbb{R}$,

$$\psi(t,x) = u(it\hbar^{-1},x) = (1+2a'it\hbar^{-1})^{-1/2} \exp\left(-\frac{a'x'^2}{1+2a'it\hbar^{-1}}\right) = \frac{\exp\left(-\frac{ax^2}{1+2a\hbar m^{-1}it}\right)}{\sqrt{1+2a\hbar m^{-1}it}}, \tag{66}$$

which, by theory, solves Eq. (61). Hence, the probability density function of the free particle is given by:

$$p(t,x) = |\psi(t,x)|^2 = \frac{1}{\sqrt{1+4a^2\hbar^2m^{-2}t^2}} \exp\left(\frac{-2ax^2}{1+4a^2\hbar^2m^{-2}t^2}\right). \tag{67}$$

## 7.2 Black-Scholes PDE

As a final example, we illustrate how the Feynman–Kac formula can be employed to solve the Black–Scholes PDE, originally introduced by Black and Scholes in [BS73]. This equation corresponds to Eq. (24) with $b(t,x) = rx$, $\sigma(t,x) = \sigma x$, $V(t,x) = r$, $f(t,x) = 0$, with $r > 0$. The solution of the $\mathbb{R}$-valued Black–Scholes PDE admits a well-known financial interpretation: it represents the theoretical price of a European-style option. In the present context we are not interested in the study of the corresponding Schrödinger equation. Rather, our aim is to highlight that this example involves an "unusual" driving process, i.e., the geometric Brownian motion.

Fix a maturity $T > 0$, a drift $r > 0$ (called *risk neutral drift*) and consider the payoff function $g$ of an European option. The Black-Scholes PDE reads, $\forall(\tau, x) \in [0,T] \times \mathbb{R}^+$:

$$\begin{cases} \partial_t u + rx\,\partial_x u + \frac{1}{2}\sigma^2 x^2\,\partial^2_{xx} u - ru = 0, & \forall (t,x) \in [0,T) \times \mathbb{R}^+, \\ u(T,x) = g(x), & \forall x \in \mathbb{R}^+, \end{cases} \tag{68}$$

so that its driving process is $dS_t = rS_t dt + \sigma S_t dB^{\mathbb{Q}}_t$, that can be interpreted as the dynamics of a risky asset under the risk-neutral probability $\mathbb{Q}$. This SDE has a well known solution, that is the geometric Brownian motion: indeed, defining $Y_t := \log(S_t)$ and applying Itô formula, we get:

$$dY_t = d(\log(S_t)) = \frac{1}{S_t}\,dS_t - \frac{1}{2S_t^2}\,d\langle S\rangle_t = \frac{1}{S_t}\left(rS_t dt + \sigma S_t dB^{\mathbb{Q}}_t\right) - \frac{1}{2S_t^2}\,\sigma^2 S_t^2 dt = \left(r - \frac{\sigma^2}{2}\right)dt + \sigma dB^{\mathbb{Q}}_t, \tag{69}$$

which is a linear SDE whose solution is $Y_t = Y_0 + (r - \frac{\sigma^2}{2})t + \sigma B^{\mathbb{Q}}_t$. So the price of the asset at each time $t \in [0,T]$ is given by:

$$S_t = S_0\,\exp\left(\left(r - \frac{\sigma^2}{2}\right)t + \sigma B^{\mathbb{Q}}_t\right). \tag{70}$$

By Feynman-Kac formula, we deduce the probabilistic characterization of the solution to the Black-Scholes PDE: $\forall(\tau, x) \in [0,T] \times \mathbb{R}$, it holds:

$$w(\tau,x) = \mathbb{E}^{\mathbb{Q}}\left[g(S_T)e^{-r(T-\tau)}\middle| S_\tau = x\right], \qquad S_T = S_\tau\, e^{\left(r - \frac{\sigma^2}{2}\right)(T-\tau) + \sigma B^{\mathbb{Q}}_{T-\tau}}, \tag{71}$$

which becomes, after time-inversion: $\forall(s,x) \in [0,T] \times \mathbb{R}$,

$$u(s,x) = w(T-s,x) = e^{-rs}\,\mathbb{E}^{\mathbb{Q}}\left[g\left(xe^{\left(r-\sigma^2/2\right)s + \sigma B^{\mathbb{Q}}_s}\right)\right]. \tag{72}$$

Using that $B^{\mathbb{Q}}_s \sim \mathcal{N}(0,s)$ under $\mathbb{Q}$, we can compute explicitly $u(s,x)$ just by integrating against a Gaussian distribution. Here, $u(s,x)$ must be interpreted as the arbitrage-free and risk-neutral price of the option as function of the stock price $x$ and of the time $s$, given the fixed maturity $T$.

We now use Eq. (72) to price an European call with maturity $T$ and strike $K > 0$ under Black-Scholes model. The payoff function is $g(x) := (x-K)^+ = \max(x-K, 0)$; hence, denoting with $\Pi^{\text{Call}}_t(T,K)$ the price of the option at time $t \in [0,T]$ with fixed maturity $T$ and strike $K$, we have$\Pi^{\text{Call}}_t(T,K) = u(T-t, S_t)$, with:

$$u(s,x) = e^{-rs}\,\mathbb{E}^{\mathbb{Q}}\left[\left(xe^{\left(r-\sigma^2/2\right)s + \sigma B^{\mathbb{Q}}_s} - K\right)^+\right] = e^{-rs}\int_{-\infty}^{+\infty}\left(xe^{\left(r-\sigma^2/2\right)s + \sigma y} - K\right)^+ \frac{1}{\sqrt{2\pi s}} e^{-\frac{y^2}{2s}}\,dy. \tag{73}$$

The integrand is non-null for $y \geq \alpha := \frac{1}{\sigma}\left(\log\frac{K}{x} - rs + \frac{\sigma^2 s}{2}\right)$, so we can write:

$$u(s,x) = e^{-rs}\left(\frac{xe^{(r-\sigma^2/2)s}}{\sqrt{2\pi s}}\int_\alpha^{+\infty} e^{\sigma y - \frac{y^2}{2s}}\,dy - \frac{K}{\sqrt{2\pi s}}\int_\alpha^{+\infty} e^{-\frac{y^2}{2s}}\,dy\right). \tag{74}$$

For the first integral, complete the square and apply the change of variable $z := y/\sqrt{s} - \sqrt{s}\sigma$; for the second integral, apply the change of variable $z := y/\sqrt{s}$. This allows us to recover integrals of Gaussian distributions, namely:

$$u(s,x) = e^{-rs}\left(xe^{rs}\int_{\frac{\alpha}{\sqrt{s}} - \sqrt{s}\sigma}^{+\infty}\frac{e^{-z^2/2}}{\sqrt{2\pi}}dz - K\int_{\frac{\alpha}{\sqrt{s}}}^{+\infty}\frac{e^{-z^2/2}}{\sqrt{2\pi}}dz\right) = x\,\Phi\left(\sqrt{s}\sigma - \frac{\alpha}{\sqrt{s}}\right) - e^{-rs}K\,\Phi\left(-\frac{\alpha}{\sqrt{s}}\right), \tag{75}$$

where $\Phi$ is the standard Gaussian cumulative distribution function. Hence,

$$\Pi^{\text{Call}}_t(T,K) = u(T-t, S_t) = S_t\,\Phi(d_1) - e^{-rs}K\,\Phi(d_2), \tag{76}$$

with

$$d_1 := \frac{\log\frac{S_t}{K} + \left(r + \frac{\sigma^2}{2}\right)(T-t)}{\sigma\sqrt{T-t}}, \qquad d_2 := \frac{\log\frac{S_t}{K} + \left(r - \frac{\sigma^2}{2}\right)(T-t)}{\sigma\sqrt{T-t}} = d_1 - \sigma\sqrt{T-t}. \tag{77}$$